

A Comparison of Heuristic and Metaheuristic Methods for solving the Traveling Salesman Problem

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Abstract—The traveling salesman problem (TSP) is an NP hard combinatorial optimization problem, that asks for the shortest possible route that visits a set of cities and returns to the starting city. Solving TSPs exactly using brute-force methods quickly becomes computationally infeasible as the number of cities increases. However, there exists a number of heuristic and metaheuristic algorithms, that can find reasonably good solutions in a much shorter time. This paper presents a comparison of five such methods, namely: Simulated Annealing, Tabu Search, 2-Opt, Iterative Local Search and a Genetic Algorithm. Algorithms are evaluated using benchmark TSP instances of sizes $n = [10, 25, 50, 100, 200, 500]$ with $k = 30$ instances each. We perform this experiment with two types of instances: one with uniformly sampled cities and one with clustered cities. They are then compared based on the solution quality measured as the gap to the optimal solution obtained by the Concorde TSP solver, and the time required to find the solution. The goal of this paper is to compare the strengths and weaknesses of these algorithms, that are the basis of many state of the art methods for solving TSPs and provide a starting point for further research into this area.

I. Introduction

II. Identified problem

The traveling salesman problem is a solved

III. Focus of this Work

IV. Thesis Organization

V. Related Work

VI. Materials and Methods

VII. Experiments and Results

VIII. Discussion

IX. Conclusion

Wollt ihr die Referenzen manuell oder mit z.B. Zotero machen?

References

- [1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.

X. The First Appendix